Week 9 — Activity: Stefan Problems and Cooling of Magma Bodies

Geophysics 3A: Potential Fields & Geothermics

May 1, 2026

Purpose

To apply transient heat flow concepts to the crystallization and cooling of magma bodies within the crust.

Physical Problem

A magma body emplaced into cooler host rock cools conductively while releasing latent heat at the solidification front.

Key Feature

The solid–liquid boundary migrates with time, making this a Stefan problem.

Tasks

1. Explain how latent heat affects the cooling rate of a magma body.
2. Compare expected cooling timescales for a narrow dike and a thick sill.
3. Identify geological situations in which analytic Stefan solutions are likely to fail.

Geological Extensions

- Magma convection.
- Episodic recharge of magma.
- Crystal-rich mush zones.

Key Insight

Geometry and diffusivity often dominate cooling timescales more strongly than radiogenic heat production.